

Problem 1

Heterocyclic chemistry is one of the most important fields of organic chemistry and biochemistry. Approximately 55% of publications in organic chemistry are related to the field, and the number of heterocyclic compounds recently found is far more than that of homocyclic compounds. The five-membered ring compounds with two heteroatoms are often present in many substances that are important for life. For example, imidazole ring is present in the essential amino acid, histidine, and its decarboxylation product, histamine. Histidine residues are found at the active sites of ribonuclease and of several other enzymes and play a vital part in the structure and binding functions of hemoglobin. Several drugs are based on the imidazole ring such as nitroimidazole, cimetidine, azomicin, metronidazole, medazolam.

1. Draw the structures of 1,3-diazole (imidazole, $C_3H_4N_2$), imidazol-1-ide anion, imidazolium cation, 1,3-oxazole (oxazole, C_3H_3NO) and 1,3-thiazole (thiazole, C_3H_3NS). Which structure(s) can be considered aromatic?

2. Arrange imidazole, 1,3-oxazole and 1,3-thiazole in decreasing order of melting and boiling points and justify your order.

3. Using structural formulae, write down equations for the ionization of imidazole, oxazole, and thiazole in water. Arrange the substances in decreasing order of base strength and justify your answer.

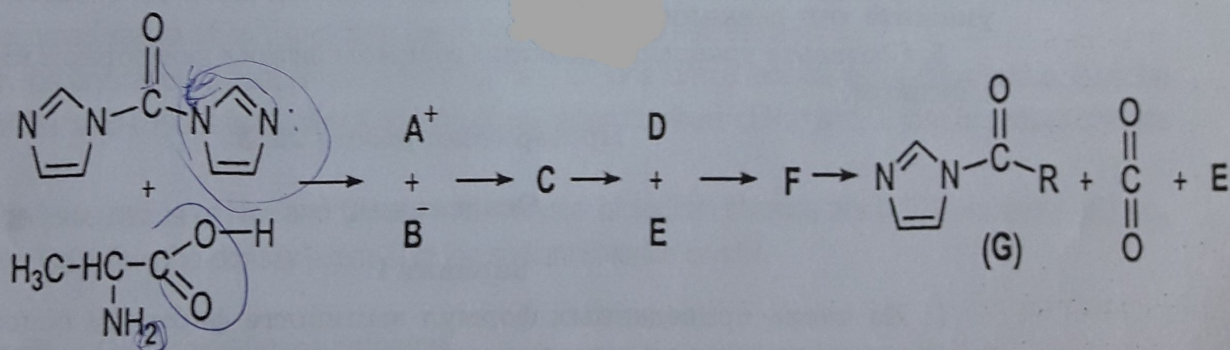
4. Propose a reaction mechanism showing the catalytic behavior of imidazole in hydrolyzing $RCOOR'$ without a participation of OH^- . Justify this behavior based on the structure of imidazole. 5. Propose a reaction mechanism for the formation of 1,1'-carbonyldiimidazole ($C_7H_6N_4O$, CDI) from imidazole and phosgen ($COCl_2$).

6. Explain why the $C=O$ stretching frequency in 1,1'-carbonyldiimidazole is 100 cm^{-1} higher than that of 1,1'-carbonyldipyrrolidine ($CO(C_4H_8N)_2$).

7. Write down reaction equations for the preparation of CDI (a) using a mixture of 4 mol imidazole and 1 mol phosgene and (b) using a mixture of 2 mol imidazole, 1 mol phosgene, and 2 mol $NaOH$. Explain why reaction (a) is preferable.

8. CDI is often used for the activation of carbonyl group for the coupling of amino acids in peptide synthesis.

8.1 Use curly arrow mechanisms to complete the scheme below, showing the formation of the active compound G from CDI and Alanine.

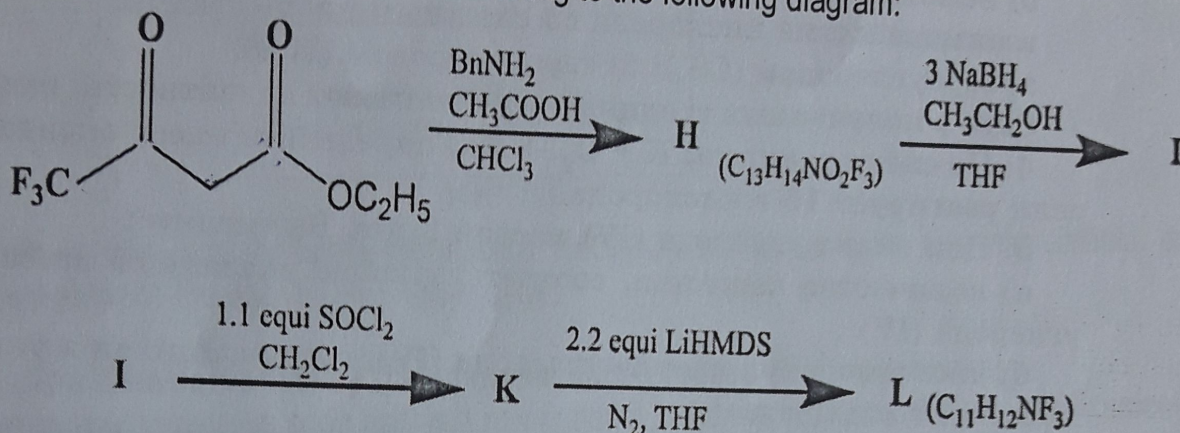


8.2 Propose a reaction mechanism for the formation of dipeptide Ala-Gly from G and Glycine.

Problem 2

Heterocyclic compounds containing the CF_3 group are interesting targets of advanced researches to generate biologically active compounds. It is quite difficult to bring the trifluoromethyl group into saturated heterocyclic structures, especially heterocycles containing nitrogen. However, these heterocycles bearing CF_3 group have several promising applications. Therefore, studies of such compounds have been carried out by many chemists.

Trifluoroacetaldehyde (A) was treated with ethanol and then refluxed with benzylamine in toluene to afford compound B ($C_9H_8NF_3$). The reaction of compound B with ethyl diazoacetate in diethyl ether with borontrifluoride etherate ($BF_3 \cdot Et_2O$) as a catalyst at $-78^\circ C$ for 4 hours provided compound C ($C_{13}H_{14}NO_2F_3$). The reduction of compound C by $LiAlH_4$ in THF at room temperature for 2 hours formed compound D. Then, D reacts with hydrogen in the presence of $Pd(OH)_2$ as a catalyst in CH_2Cl_2 at room temperature for 60 hours to obtain E ($C_4H_6NOF_3$). Compound E was then allowed to react with 2 equivalents of tosyl chloride (TsCl) in dicloethane with the catalytic Et_3N and amount of 4 dimethylaminopyridine (DMAP). The reaction was carried out at room temperature for 2 hours, then 3 hours of reflux, to furnish compound F. F reacted with 1.2 equivalents of phenol in the presence of K_2CO_3 in DMF to form compound G ($C_{17}H_{16}NSO_3F_3$). A derivative of G can be synthesized according to the following diagram:



1. Complete synthetic schemes from B to L.
2. Write reaction mechanisms from B to C.
3. Write reaction mechanisms from F to G.

Note:

$BnNH_2$: benzylamine, $C_6H_5NH_2$; Bn: Benzyl, $C_6H_5CH_2$; THF: tetrahydrofuran LiHMDS: Lithium bis(trimethylsilyl)amide, Et: C_2H_5 ; Equi: equimolarity

Problem 3

Crystal structure of alkali metal halide In crystals of ionic compounds, cations are generally arranged in the interstices of the closest packed lattice of anions. The structure of an ionic crystal such as sodium chloride becomes stable when the cations are in contact with the nearest anions.

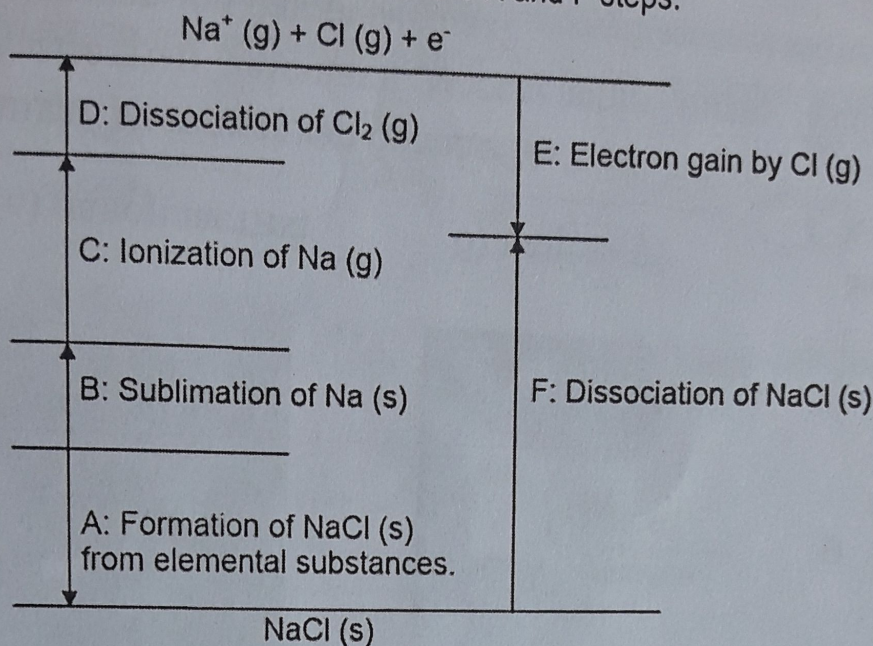
a) In the crystal of sodium chloride, both Na^+ and Cl^- ions form a face-centered cubic lattice. Give the numbers of Na^+ and Cl^- ions in a unit cell and the coordination numbers of Na^+ and Cl^- ions in sodium chloride crystal.

b) The ionic radius of Na^+ and Cl^- ions in the crystal of sodium chloride are 0.102 nm and 0.181 nm, respectively. Calculate the density [$kg\ m^{-3}$] of the sodium chloride crystal.

Born-Haber cycle and lattice enthalpy

In ionic inorganic compounds such as sodium chloride, the heat of lattice formation from gaseous ions is very high, and the contribution of the change in entropy is small. Therefore, the lattice formation enthalpy is estimated from enthalpy data by using a Born-Haber cycle.

c) The figure below shows the Born-Haber cycle of NaCl. The labels "g" and "s" represent "gas" and "solid" states respectively. Show chemical equations in the A and F steps.

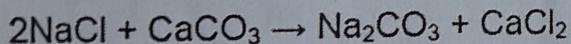


d) Calculate the enthalpy of the lattice formation of NaCl [kJ mol^{-1}] by using the following enthalpy data of the respective steps in the above Born-Haber cycle.

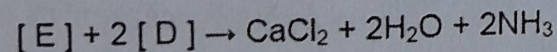
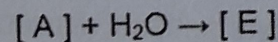
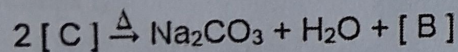
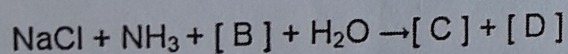
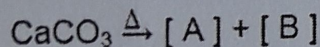
Synthesis of sodium carbonate by the ammonia-soda process (Solvay process)

Sodium carbonate (anhydrous soda ash) is a raw material in the manufacture of glass, medicaments, alkaline detergents, etc.

e) The total chemical reaction in the ammonia-soda process is represented as follows:



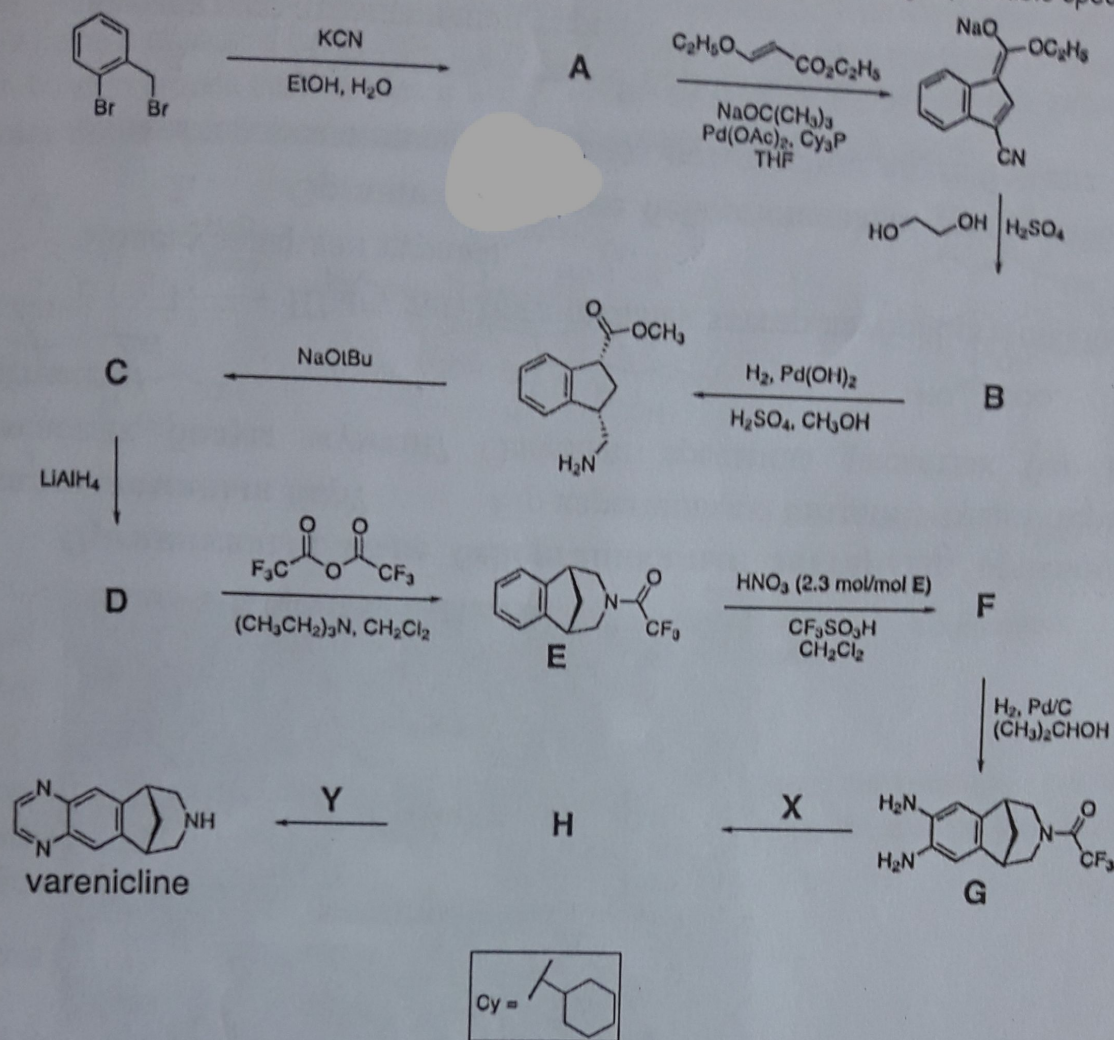
This reaction between sodium chloride and calcium carbonate does not proceed directly. The process comprises the following five reactions involving ammonia:



where Δ represents applying heat treatment. Insert the chemical formulas of the appropriate compounds in the blank spaces [A]–[E] in the above reactions.

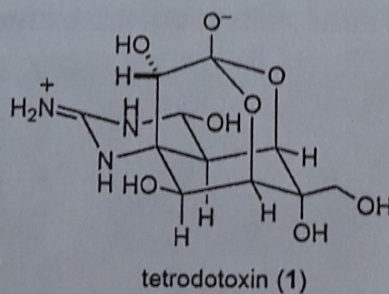
Problem 4

Varenicline has been developed as an oral treatment for smoking addiction and can be synthesized by the route shown below. All compounds indicated by a letter (A – H) are uncharged, isolable species.

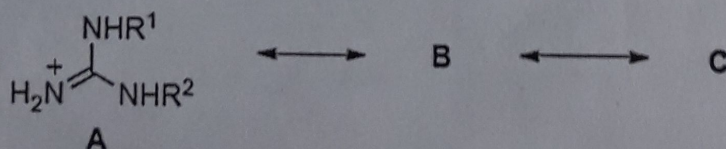


Problem 5

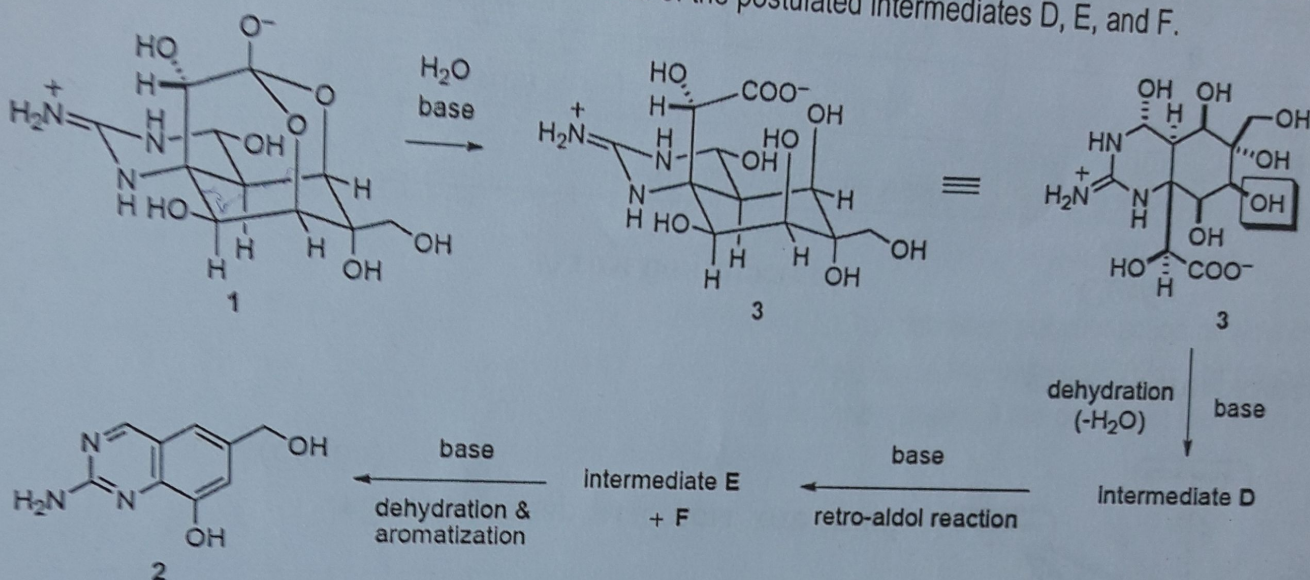
Certain varieties of puffer fish, Fugu in Japanese, are highly prized as foods in Japan. Since the viscera (especially ovaries and livers) of the fish contain a potent toxin (tetrodotoxin), food poisoning often results from its ingestion. Studies on tetrodotoxin (1) have been performed from the beginning in the 20th century; its chemical structure was elucidated in 1964.



a) The guanidine group in tetrodotoxin exhibits strong basicity. The guanidinium ion resulting from protonation on the guanidine group is stabilized by the existence of the following resonance. Draw two resonance structures B and C.

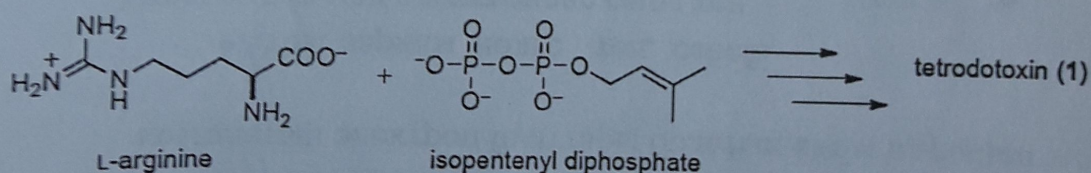


b) Many derivatization reactions were performed in structure studies of tetrodotoxin. Treatment of tetrodotoxin (1) with ethanolic potassium hydroxide upon heating afforded quinazoline derivative 2, which provided an insight into the nature of the fundamental skeleton of tetrodotoxin. The reaction mechanism can be described as follows. First, tetrodotoxin is hydrolyzed into carboxylate 3. Then the hydroxyl group highlighted with a frame is eliminated by the base to give intermediate D. A retroaldol reaction of D cleaves a carbon-carbon bond to provide intermediates E and F. Finally, dehydration and aromatization from E produce quinazoline derivative 2. Draw structures of the postulated intermediates D, E, and F.

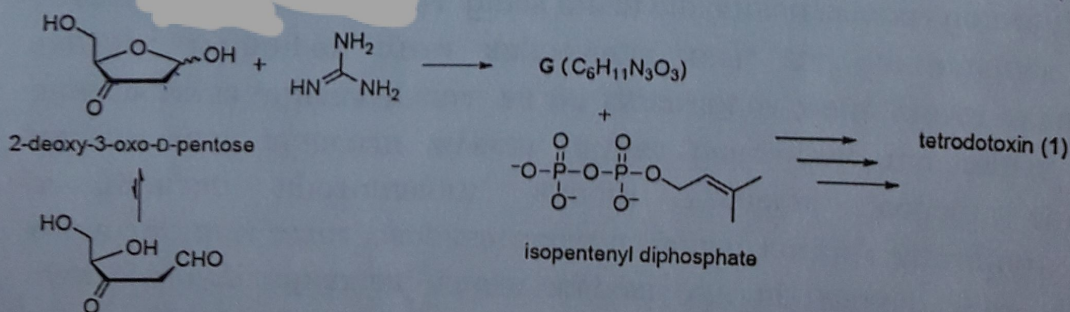


c) Although biosynthesis of tetrodotoxin still remains to be clarified, it is proposed that tetrodotoxin may be biologically synthesized from L-arginine and isopentenyl diphosphate. Among the carbons included in tetrodotoxin, circle a

II the carbons that are expected to be of L-arginine origin.



d) In the 1990s, an alternative biosynthetic pathway of tetrodotoxin was proposed. Condensation between 2-deoxy-3-oxo-D-pentose and guanidine provides intermediate G with cyclic guanidine moiety (molecular formula C₆H₁₁N₃O₃). Tetrodotoxin may be biologically synthesized from intermediate G and isopentenyl diphosphate. Draw a structure of the postulated intermediate G showing the stereochemistry.



Problem 6

- a) The productivity of the tubular ethylene polymerization reactor at 170 MPa is 6,000 kg of polyethylene per hour. The reactor is a pipe with a diameter of 60 mm and has a length of 1000 m. Determine the volumetric feed rate of ethylene (at the indicated pressure and average 47 gas temperature of 190 °C), if the conversion rate of ethylene is 12.5%.
- b) The mass consumption of styrene in the emulsion polymerization plant is 1300 kg/h, the volume fraction of styrene in the mixture entering the polymerization is 34%. Determine the number of reactors for a continuous process if the residence time of the mixture in the cascade of reactors is 4 hours, the volume of each reactor is 5 m³, the reactor fill factor is 0.84. The density of styrene is 906 kg / m³.
- c) The mass fraction of PVC in the latex obtained by the emulsion polymerization of vinyl chloride is 42%. The conversion rate of the monomer is 95%, the capacity of the unit is 1500 kg / hr for polyvinylchloride. Determine the volume ratio of vinyl chloride: the aqueous phase, if the density of the emulsion is 1120 kg / m³, and the density of vinyl chloride is 973 kg / m³.